

Excellent. You now have:

- ✓ Product Blueprint
- ✓ DDD (Database Design Document)
- ✓ BASD (Backend API Specification Document)

The next document should be:

Backend Data Flow Specification (BDFS)

ScrapFlow AI – Version 1.0

This is the document that connects:

Database

+

APIs

+

Business Rules

and answers:

"What exactly happens inside the backend when a user performs an action?"

Why BDFS Is Important

DDD tells us:

What data exists

BASD tells us:

What APIs exist

BDFS tells us:

How data moves

How business logic works

How services interact

Sections of BDFS

1. Authentication Flow

User Login



Validate Email



Validate Password



Generate JWT



Return Token

2. Batch Creation Flow

Create Batch Request



Validate Input



Generate Batch Record



Store Collection Cost



Store Fuel Cost



Save Batch



Return Success

3. Material Addition Flow

Select Batch



Add Material



Validate Weight



Store Material



Update Inventory



Return Success

4. Inventory Update Flow

Material Created



Find Material Type



Check Existing Inventory



Update Weight



Save Changes

Example:

Copper Existing = 100kg

New Copper = 20kg

Inventory = 120kg

5. Sales Flow

Create Sale



Validate Buyer



Validate Inventory



Calculate Revenue



Store Sale



Reduce Inventory



Update Dashboard

6. Expense Flow

Create Expense



Validate Amount



Store Expense



Update Profit Calculation

7. Profit Engine Flow

Revenue



Get Expenses



Calculate Profit



Update Dashboard

Formula:

Profit =

Revenue

-

(Collection Cost

+ Fuel Cost

+ Expenses)

8. Dashboard Data Pipeline

Sales Table



Expenses Table



Inventory Table



Profit Engine



Dashboard API



Frontend

9. Report Generation Flow

User Requests Report



Fetch Data



Generate Report



Create PDF/Excel



Download

10. Service Layer Design

This is extremely important.

routes/



services/



repositories/



database/

Example:

Sales Route



Sales Service



Sales Repository



MySQL

11. Transaction Flow

Critical for data integrity.

Example:

Create Sale



Update Inventory



Store Sale

If inventory update fails:

Rollback Sale

Use MySQL transactions.

12. Error Handling Flow

Request



Validation



Service Layer



Database



Response

Error Example:

```
{  
  "success": false,  
  "message": "Inventory insufficient"  
}
```

13. Future AI Flow

Voice Note

Voice Recording



Upload Audio



Store Audio



Transcription Service



Store Text

AI Assistant

Question



AI Service



Database Query



Generate Answer



Return Response

After BDFS

The next document should NOT be more backend planning.

The next document should be:

UI/UX Specification Document (UXSD)

Contains:

Dashboard Screen

Collections Screen

Inventory Screen

Sales Screen

Reports Screen

Navigation Design

Component Design

Color System

Mobile Layout